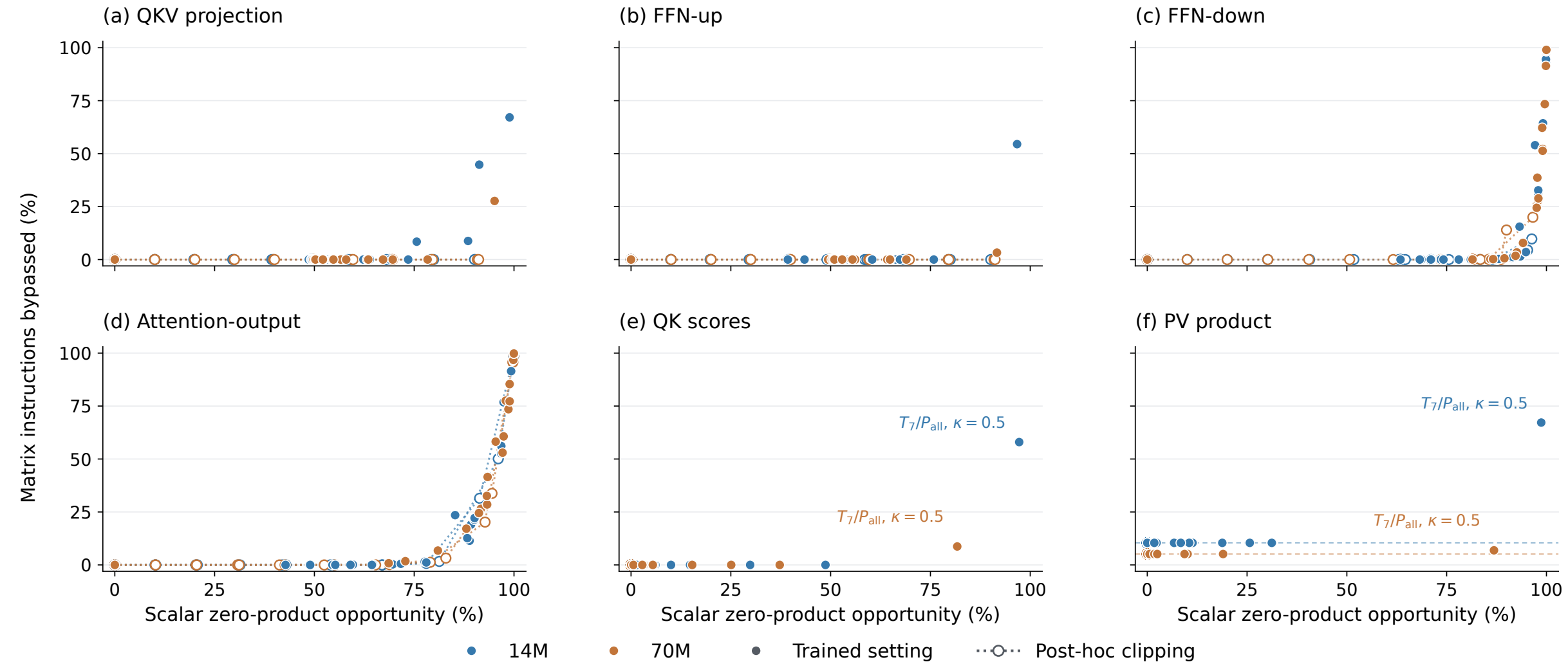


Scalar Sparsity and Matrix-Instruction Bypass



Scalar opportunity: BF16 activation-derived lower bound. Bypass includes scalar substitution and attention padding.

PV dashed guides: Base model bypass = 10.42% (14M), 5.15% (70M).